

Tamara AiQ Fluency Assessment

Phase 1 — what we have built, and how scoring works

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1. Executive summary

AiQ is a conversational AI-fluency assessment: instead of a quiz, the participant spends about ten minutes in a chat where an AI interviewer walks them through one realistic work scenario from their own job family, presses on what goes wrong, and asks what standard they would hold others to. A scoring pass then reads the full transcript and produces a 0–100 AiQ score, a maturity band, six dimension scores each backed by the participant's own verbatim quotes, and a personal coaching report with a downloadable PDF.

The product is now pilot-ready for the three chosen job families — product & engineering, go-to-market, and care & operations — across all four seniority levels (12 combinations). Every scoring claim below has been validated with deliberate strong-vs-weak transcript tests against the live model, not assumed.

2. What is built

2.1 Participant experience

- **Onboarding** — a branded home page and a six-step flow: what brings you here, seniority and area (this drives everything downstream), an honest self-rating, a “place your bet” score guess compared against the real result at the end, and house rules.
- **Scenario chat** — a five-phase structured interview: context anchor → main scenario → complication (something goes wrong) → standards (the bar they would set for others) → wrap-up. The server owns the phases and the scenario facts; the language model only phrases the next question, so the flow can never wander off-script.
- **Conversational quality** — follow-up questions are generated from what the participant actually said, so two people never get identical canned questions. Clarification requests (“what do you mean?”) are answered and the question re-asked without burning progress. Reactions are filtered so the interviewer never grades an answer mid-chat — this keeps the assessment neutral.
- **Session safety** — sessions survive page reloads, a progress bar shows which part of the conversation they are in, and the end-session dialog is gated: it will not let someone finish with too little signal to score fairly.

2.2 Pilot scenario matrix (3 families × 4 levels)

Each family has its own realistic scenario. Seniority does not change which story — it changes altitude: ICs are asked for their own prompts, edits, and checks; executives for governance and how they set expectations. Product/engineering and care/operations get a second, strategic scenario for head-of and executive tiers, because one story cannot credibly stretch from “draft a spec” to “set function-wide policy”. Go-to-market auto-adapts (launch copy / sales outreach / brand campaign) from the participant's first answer.

Family	ic / people manager	head of / executive
Product & engineering	Spec due Friday — AI drafts a payments spec needing compliance sign-off; real account numbers get pasted into a shared chat	Build vs. buy — AI code-review tool decision; an engineer skips the pilot process with production code

Family	ic / people manager	head of / executive
Go-to-market	Launch copy / sales outreach / brand campaign (auto-selected) — AI-drafted customer-facing content; an unapproved claim slips into the draft	Same scenario, asked at portfolio altitude (claims governance, sign-off ownership)
Care & operations	Angry VIP customer — an AI apology offers a refund bigger than policy allows, one click from sending	AI rollout across the ops floor — mixed pilot data: handling time down in one queue, escalations quietly rising in another

2.3 Engineering under the hood

- Flask + SQLite backend with rate limiting, session expiry, resume-on-reload, and an idempotent scoring endpoint (a browser timeout cannot lose a finished score).
- LLM abstraction with OpenAI as the pilot backend and automatic local-model fallback if quota is hit mid-session.
- A simulation harness replays 16 personas (strong, terse, vague-enthusiast, clarification-heavy) through the full pilot matrix and fails CI if conversation quality regresses; plus a scoring-validation harness (section 4) and three unit-test suites — all green.

3. The scoring logic

Scoring is designed around one principle: a score must be defensible. Every number is anchored to observable behavior and backed by the participant's own words, and nothing about the composite is left to the model's discretion.

3.1 Six dimensions, scored 0–10 against behavioral anchors

Dim	Measures	Low (0–3)	Mid (4–7)	High (8–10)
D1 Awareness	Where AI is (and isn't) worth applying	≤1 tool named in passing	2+ tools mapped to real tasks	Maps opportunity across the function; can name where they chose not to use AI
D2 Prompts & comms	Directing AI effectively	Vague one-shot prompting	Gives context, iterates on weak output	Repeatable approach (templates, structured context) per task type
D3 Critical judgment	Verifying output, when to trust	Takes output at face value	Caught a wrong output before use (concrete case)	Standing verification habit; knows where AI is reliable vs not
D4 Workflows & org	How AI is embedded in real processes	Ad hoc, individual use	Embedded in a workflow with an owner	Redesigned a process around AI with explicit handoffs
D5 Output craft	Quality of AI-assisted work others rely on	Ships AI output unedited	Edits for tone/accuracy, rough bar	Explicit, enforced output standard
D6 Risk & responsible use	Data boundaries, escalation	No mention of boundaries	Names a concrete boundary + who to escalate to	Proactive, consistent data-handling practice

The assessor model must place each dimension in a band first, then fine-tune in 0.5 steps. Job title, confidence, and enthusiasm are explicitly excluded as evidence — only observable behavior in the transcript counts.

3.2 No score without a quote — enforced in code, not just prompted

The pipeline runs in two passes. An evidence-extraction pass first pulls structured signals (tools in use, prompt behavior, verification, workflow ownership, output bar, risk floor) with supporting quotes. The scoring pass then produces the six dimension scores, each with one to two verbatim quotes from the participant's own turns.

- **Quote grounding:** every returned quote is checked against the actual transcript (normalized substring match). A paraphrased or invented quote is silently dropped — the model cannot fabricate evidence.
- **The cap:** if no grounded quote survives for a dimension, its score is capped to the low band (≤ 3) in code. A confident-sounding transcript cannot earn a high score that no quote backs up. We watched this work live: an executive session that never discussed output quality scored 3/10 on D5 — correctly.
- **Low-effort gate:** a whole-session effort signal (share of thin replies, average words) blocks scoring entirely when someone answers “idk / not sure / yes” throughout — no confident-looking score built on nothing.

3.3 Weights: same rubric for everyone, weighted per profile

Every dimension is judged the same way, but its contribution to the composite depends on the participant's level and family: a base weighting plus a level delta plus a family delta, renormalized to sum to 1.0. All twelve pilot combinations get a distinct, computed profile — ICs are weighted toward hands-on prompting and judgment, executives toward workflows and risk governance:

Profile	D1	D2	D3	D4	D5	D6
IC · product & engineering	0.19	0.18	0.23	0.20	0.11	0.09
Executive · product & engineering	0.14	0.13	0.19	0.24	0.10	0.20
IC · go-to-market	0.20	0.16	0.22	0.17	0.15	0.10
Executive · go-to-market	0.15	0.11	0.18	0.21	0.14	0.21
IC · care & operations	0.19	0.16	0.23	0.20	0.12	0.10
Executive · care & operations	0.14	0.11	0.19	0.24	0.11	0.21

3.4 Composite and band — computed, never trusted from the model

The final number is recomputed in code from the dimension scores: $\text{AiQ (0–100)} = 10 \times \sum (\text{weight}_i \times D_i)$. The maturity band is then derived deterministically from that composite: AiQ1 (0–25), AiQ2 (26–55), AiQ3 (56–80), AiQ4 (81–100). The model's own arithmetic and band guess are discarded — this closed a real bug found during validation, where a 25-point and an 80-point transcript could both be labelled “AiQ3”.

Finally, all participant-facing text is sanitized: internal assessor phrasing (“the user failed to provide...”) is replaced with neutral, second-person coaching language before it reaches the report.

4. Validation — confirmed, not assumed

Before any real participant, we ran deliberate sanity checks per family: one transcript written to look strong and one written weak, at both the IC and executive ends, through the real evidence + scoring pipeline against the live model. All checks pass:

Family / level	Strong	Weak	Separation	Bands
Product & engineering · IC	78.0	34.1	43.9	AiQ3 vs AiQ2
Product & engineering · Executive	73.7	28.3	45.4	AiQ3 vs AiQ2
Go-to-market · IC	75.9	37.5	38.4	AiQ3 vs AiQ2
Go-to-market · Executive	72.2	35.5	36.7	AiQ3 vs AiQ2
Care & operations · IC	72.0	36.0	36.0	AiQ3 vs AiQ2
Care & operations · Executive	82.3	26.6	55.7	AiQ4 vs AiQ2

- Strong and weak transcripts separate by 36–56 points everywhere (threshold: ≥ 15) — clearly separated, not clustered.
- Every evidence quote returned was verified to exist verbatim in the transcript — zero fabricated quotes across all runs.
- Internal consistency holds: weak transcripts with no risk-awareness content never scored above the low band on D6.
- All 12 level × family combinations resolve to the intended scenario (automated matrix check), and 16 persona simulations reach every conversation phase — including the standards phase, which a turn-budget bug had previously made unreachable (fixed and regression-tested).
- One full live end-to-end session (executive, care & operations) was run through the real UI: correct strategic scenario, all five phases, scored 73.1 / AiQ3 with grounded quotes on the report.

5. Known limits and what comes next

- Exec-tier scenarios elicit output-craft (D5) signal weakly — the live exec session correctly floored D5 at 3 because the topic never came up. We will watch the “all six dimensions signalled $\geq 80\%$ ” exit metric closely in the pilot.
- Whether go-to-market’s single auto-adapting scenario holds across all four levels is a question the pilot itself will answer.
- Deliberately out of Phase 1: real-time per-message scoring, persona classification, org-level dashboards, SSO.

Exit criteria for the pilot: $\geq 70\%$ completion, 8–12 minute median duration, $\geq 80\%$ of sessions with all six dimensions signalled, $< 5\%$ scoring failures, and the validation suite green per family — the last of which is already met.